

Centient Deliverable 1

QA Readiness Guide

Developer-to-QA handoff for GitHub issue #13 - Instant USDC Reward Rail on Stellar

Document version	1.0 (2026-09-08)
Purpose	Give QA every developer-owned fact, control, and evidence location needed to execute the 28 cases in Centient_D1_Test_Cases.docx freely and consistently. This is a readiness handoff, not an executed QA report.
Developer-readiness issue	#13 - Prepare Deliverable 1 for unrestricted QA execution
QA execution issue	#80 - Deliverable 1 QA execution and test-case refinement
Epic	#4 - Deliverable 1 - Instant USDC Reward Rail on Stellar
Network	Stellar Testnet only. No mainnet transaction is authorized at any point in this guide.
Frozen build (full SHA)	fa36cf4663fecea6137dcd4e73b136baa49d34f0
Deployment timestamp	2026-09-08T11:06:29.346Z
Build URL	https://centient.work (Railway public domain: https://web-production-05914d.up.railway.app)
Approval signal	QA applies the QA: PASSED label on #80. No comment, review link, issue closure, or Epic update is required from QA.

Redaction statement. This document contains only public Stellar Testnet identifiers, transaction hashes, variable names, and public URLs. It deliberately contains no seeds, private keys, HMAC secrets, database URLs, webhook URLs, access tokens, passwords, signed pending XDRs, or raw environment dumps. Every QA evidence artifact must follow the same rule (section 11).

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1. Executive readiness assessment

Conclusion. The QA document is strong. Its 28 cases are well structured, correctly scoped to Stellar Testnet, and cover every Deliverable 1 issue. It needs only targeted tweaking, which #80 owns. **Unrestricted QA is not ready yet:** the developer-owned blockers listed below must be resolved first, because without them several cases cannot be executed without code changes, secrets, or guesswork. The blockers are concrete and bounded; none requires redesigning the QA suite.

Developer-owned blockers (must be closed before #80 begins unrestricted execution)

ID	Priority	Blocker	What resolves it	Cases affected
B1	Blocked P0	Scheduler and alert channel are not configured on the live web service: CRON_SECRET and DISCORD_WEBHOOK_URL are absent. The wallet-health, payout-retry, and reserve-refill cron routes cannot be invoked, and Discord alerts cannot be observed.	Set both variables on the web service (values stay in the secret store), provision an external scheduler that calls the three cron routes, redeploy, and give QA a test Discord channel to watch.	TC-021, TC-022, TC-023, TC-024, TC-026 (alert half), TC-017 (breach alert)
B2	Blocked P1	Runbook and evidence mismatch. The multisig, spike, and payout-service runbooks document payout account GC5UOKLU . . . TTR4A6, but the deployed account is GCP34RIT . . . WK4BU0.	Update the three runbooks to the live account, signers, and transactions, or label the existing sections clearly as historical and add a current-deployment section. Section 4 has the full table.	TC-001, TC-003, TC-027
B3	Blocked P1	Cold-reserve policy variables (STELLAR_COLD_RESERVE_ACCOUNT, cold signer publics, trigger/target/retain units) are absent from the live web service, so the reserve-refill detection path is not live.	Set the public cold identities and the three policy unit values on the web service; confirm POST /api/cron/reserve-refill returns a plan; record the policy values in #80.	TC-021, TC-020 (detection half)
B4	Blocked P1	Co-signer database role is unproven. The cosigner service holds both COSIGNER_DATABASE_URL and a generic DATABASE_URL. Variable presence cannot show whether the role is read-only, and the runbook says the co-signer must not carry the application's read-write URL.	Run the runbook permission probes as the co-signer role and attach a redacted transcript showing required reads succeed and writes to decision data fail. Remove or justify DATABASE_URL on the cosigner service.	TC-016, TC-013 (indirect)
B5	Caveat P1	Provenance of the live two-signature payout 5083dd72a16b . . . is not yet recorded. The only documented payout-service proof used the gated local co-signer, which does not prove the deployed remote co-signer path.	Record which co-signer produced the second signature on the live payout. If it was the local signer, run one payout through the deployed cosigner service and record its hash.	TC-004, TC-013
B6	Blocked P0	QA-safe fixtures and fault-injection controls do not yet exist as documented, access-controlled, testnet-only mechanisms (ambiguous Horizon response, post-broadcast persistence failure, concurrency runner, cap-usage seeding, cooldown control, health-source outage, single-signer attempt).	Deliver the controls in section 7 with exact commands or endpoints, then reflect them in the DOCX revisions under #80.	TC-002, TC-006 to TC-015, TC-017 to TC-020, TC-022, TC-023
B7	Blocked P1	The legacy PAYOUT_PRIVATE_KEY variable on the web service should be rotated before QA handoff. The variable belongs to the retired EVM path and is not used by the Stellar rail.	Rotate the value (requires explicit owner authorization) or remove the variable if the legacy path is fully retired. Do not paste any value into evidence.	TC-027 (evidence hygiene)

Handoff items owned by QA through #80 (not developer blockers)

ID	Owner	Item	Action
H1	QA-owned	The DOCX still uses obsolete wording: the section title #13 Final manual QA gate and the verdict QA PASSED <full-sha> in TC-027, TC-028, and the execution summary.	Revise under #80 so that the approval action is the QA : PASSED label on #80 and the developer-readiness gate is described as #13. Keep test IDs unchanged.
H2	QA-owned	Cases that say "if supported", "where applicable", or "controlled ability" (TC-009, TC-010, TC-011, TC-013, TC-018, TC-019, TC-023).	Replace with the exact developer-provided control from section 7 once B6 is delivered.

Accepted decisions that QA records but does not block on

ID	Status	Decision	How QA records it
A1	Accepted	The web service holds both the payout-account master credential and the ops-signer credential. The team decided not to move to isolated infrastructure for the testnet MVP, so the deployed application can produce two of the three payout signatures.	Treat as an accepted residual risk alongside ADR-0001 (section 10, R7). TC-002 still verifies that any single key is refused on-chain; it does not certify per-process key separation.
A2	Accepted	The co-signer runs as a separate service inside the same Railway project, with COSIGNER_ISOLATION_LEVEL=same-workspace. This is the amended ADR-0001 topology.	Record the isolation level in force at the tested SHA and cite ADR-0001 (section 8).

2. Testnet environment and frozen build

All facts below were verified on 2026-09-08 against the live deployment and public Horizon data. QA executes against this single environment and this single SHA. If the deployed SHA changes during the run, the developer notifies QA on #80 and affected cases are reset.

Public site	https://centient.work
Railway public domain	https://web-production-05914d.up.railway.app
Railway project	centient-work
Railway environment name	production (the environment label only; the Stellar network configuration is testnet)
Stellar network	STELLAR_NETWORK=testnet - Horizon https://horizon-testnet.stellar.org
Services	web (application, payout worker), cosigner (policy co-signer), Postgres (shared database instance)
Co-signer exposure	No public domain. Reached by the web service over same-project private networking only.
Isolation level	COSIGNER_ISOLATION_LEVEL=same-workspace
Deployed commit	fa36cf4663fecea6137dcd4e73b136baa49d34f0
Deployment succeeded at	2026-09-08T11:06:29.346Z for both web and cosigner
Application daily cap	DAILY_PAYOUT_CAP_UNITS=2000000000 (200 USDC, rolling 24 h)
Co-signer daily cap	COSIGNER_DAILY_CAP_UNITS=2000000000 (200 USDC, UTC calendar day)
Scheduler secret	CRON_SECRET absent on web (blocker B1)
Alert webhook	DISCORD_WEBHOOK_URL absent on web (blocker B1)
Cold-reserve policy on web	Absent (blocker B3)
Public wallet health	https://centient.work/api/health/wallet

Live wallet-health response at verification time

GET /api/health/wallet is unauthenticated and safe for QA to call at any time. It reports both balances live; the health monitor cron adds payout activity, cap, failure, and reserve metrics but needs blocker B1.

Account	GCP34RITQIVSLHS5T4XZRENIBUS3T7FHL3VSR24GK7HPMHGAAKWK4BU0
USDC balance	19.0000 USDC
XLM balance	9999.9999 XLM (spendable 9997.4999 XLM after a 2.5 XLM minimum balance)
USDC status	warn - below the 50 USDC warning threshold
XLM status	healthy
Overall healthy	false (driven by the USDC warning only)
Thresholds in force	USDC warn 50 / page 10; XLM warn 5 / page 2 (defaults)

The current USDC warning is a real, reproducible low-float state and is useful for TC-022. Before the payout-heavy cases (TC-004, TC-009, TC-017 to TC-019) the developer tops the float up from the Circle testnet faucet or the cold reserve so payouts are not starved. Record the balance before and after any top-up in #80.

3. Public on-chain evidence

Every identifier below is public. QA can open each link without credentials. Horizon JSON for any account is <https://horizon-testnet.stellar.org/accounts/<G...>> and for any transaction <https://horizon-testnet.stellar.org/transactions/<hash>>.

3.1 Live payout (hot) account - currently deployed

Payout account (master)	GCP34RITQIVSLHS5T4XZRENIBUS3T7FHL3VSR24GK7HPMHGAAKWK4BU0 stellar.expert account
Thresholds	low 2 / medium 2 / high 2
Master signer weight	1
Ops signer (weight 1)	GAHZKJFAWX3HXAYQFAPCJ3Y2DFHYB0CSNVBJTZRDYJXN763KIR6HZ2U6
Policy signer (weight 1)	GCAUNAS2ZHBR0HNKP32KWMEPJGMX5XJMSGNKXV3YVMRZ72SFBL3PKHIU
Effective policy	Any 2 of 3 keys authorize a payment; no single key can. Verified on Horizon 2026-09-08 (3 signers, each weight 1).
Setup (set-options) tx	e966c0a5c27cbe0253f2812d158b38f8e91513de48254923f9ecb6f4c19630fd stellar.expert transaction - 3 operations, source = payout account, 2026-09-08T06:30:17Z
Successful 1 USDC payout	5083dd72a16bfa749c6b302c293e931939acd70cc52f63586443921d60698206 stellar.expert transaction - fee-bump envelope, 2 signatures on the outer envelope and 2 on the inner payment, fee charged 200 stroops by the payout account, ledger 4565647, 2026-09-08T07:17:02Z
Balances at verification	19.0000000 USDC, 9999.9999400 XLM

3.2 USDC asset

Asset	USDC (Circle Stellar Testnet issuance), 7 decimals; 1 USDC = 10000000 units
Issuer	GBBD47IF6LWK7P7MDEVSCWR7DPUWV3NY3DTQEVFL4NAT4AQH3ZLLFLA5 stellar.expert issuer account
Faucet	Circle testnet faucet

3.3 Cold reserve account (2-of-3)

Cold account (master)	GDPGRS4P6UZZK23CKKELGLJAYTCAWPV4C7TH6Q322SF735A5H6U5XK5G stellar.expert account
Thresholds	low 2 / medium 2 / high 2; master, ops, policy each weight 1 (verified on Horizon 2026-09-08)
Cold ops signer	GDERX2QG4LY4SK6VHBX5VRKE2PFDB25ZCD30LUM4RDVAY43E0XBRZ4BC
Cold policy signer	GA0TECDRSB5HHAJAM00MUYYTD0R5HTFNSETEDNWJDXDFRBZVQM00XD6V
Current balance	6.90000000 USDC, 9999.9999500 XLM
Trustline tx	fddea73e981194f020bcf531f96140c3117f8c400915228e6901c2312b4a8c6a stellar.expert
Setup tx	05bbe397033a7984700b8d844cf1d247e04db7537fd9a0293a5bda6c1506d14a stellar.expert
Initial funding tx (hot to cold)	452fd68061ecae052ebd681ee47010adbc2e05c01c86afb9bba685a77fb1d836 stellar.expert - 10 USDC, sent through the multisig payout service from the historical payout account
Refill tx (cold to hot)	3a5969cdac22dad6646630c90cdb3ae3919a727f2abd8f663863b7e9e6b9ef3e stellar.expert - 3.1 USDC, source = cold account, 2 signatures (cold master + cold ops), fee 100 stroops, ledger 4563417
Historical refill policy	trigger 10 USDC, target 12 USDC, retained cold minimum 5 USDC (100000000 / 120000000 / 500000000 units). Not configured on the live web service - see blocker B3.

3.4 Historical evidence (superseded payout account)

These transactions are valid on-chain proofs of the transaction shapes, but they were produced from the earlier payout account and signer set. They are cited by the runbooks today and must be labelled historical (blocker B2).

Historical payout account	GC5U0KLU6J2ER0ZYYP2I23ZEF4YF42TGGRNQMMTG0JGJ7N0CH3TTR4A6 stellar.expert account
Historical ops signer	GDNL20G7XGBHTPNW4WQAT7AVLXYFHAP76DVYFMSKIZLP47KYM0LS27V3
Historical policy signer	GB6NBHA5ML3D0AQXBSDRNYVJUE6B3VPEP2BH5ZXPL5D6YZV5DJJC6IWL I
Historical setup tx	8a16d6236cbb0aa1887eba35f6b4b04e41e289532df06b1d13413551e8fc1ad5 stellar.expert
Sponsored recipient	GA4XSHGIOVGWI6ZGDIRFG3R0F0HXRDVHH5YKGFKC5C2LH0T6EB5WCQXI stellar.expert account - created with zero XLM, USDC trustline sponsored by Centient
Sponsored-trustline tx	63dba35261787b2fa05c0c1c833a02f5983c3a4e229d984a7ed1ecf3c881a71b stellar.expert
Payout spike tx (#6)	be8a6a2363401ff77f3634b8af47db88a4e2706e401319fa8ecd2412342e9154 stellar.expert - 1 USDC, fee-bump, 2 + 2 signatures, fee 200 stroops, recipient spent 0 XLM
Payout-service proof tx (#71#73)	9f6b8ae96a2d17cf3ba2fd585f1e434287bc7113d33f5ecec75cbbe53470996c stellar.expert - 0.1 USDC through the real payReward path, but with the gated local co-signer . It does not prove the deployed remote co-signer path.

4. Live deployment facts versus older repository evidence

QA will find two sets of identifiers in the repository. This table says which is which, so that evidence is tied to the deployed build and not to a superseded proof.

Item	Live deployment (verified 2026-09-08)	Repository docs at the base SHA	Resolution
Payout account	GCP34RITQIVSLHS5T4XZRENIBUS3T7 FHL3VSR24GK7HPMHGAAKW4BU0	GC5U0KLU6J2ER0ZYYP2I23ZEF4YF42TGGR NQMTG0JGJ7N0CH3TTR4A6 in stellar-multisig-runbook.md, stellar-multisig-payout-spike.md, stellar-multisig-payout-service.md	B2: update or label historical
Ops signer	GAHZKJFAWX3HXAYQFAPCJ3Y2DFHYB0 CSNVBJTZRDYJXN763KIR6HZ2U6	GDNL20G7XGBHTPNW4WQAT7AVLXYFHAP76D VYFMSKIZLP47KYMOLS27V3	B2
Policy signer	GCAUNAS2ZHBROHNKP32KWMEPJGMX5X JMSGNKXV3YVMRZ72SFBL3PKHIU	GB6NBHA5ML3D0AQXBSDRNVYJUE6B3VPEP2 BH5ZXPL5D6YZV5DJC6IWL I	B2
Set-options tx	e966c0a5c27cbe0253f2812d158b38 f8e91513de48254923f9ecb6f4c196 30fd	8a16d6236cbb0aa1887eba35f6b4b04e41 e289532df06b1d13413551e8fc1ad5	B2
Two-signature payout	5083dd72a16bfa749c6b302c293e93 1939acd70cc52f63586443921d6069 8206 (1 USDC)	be8a6a2363401ff77f3634b8af47db88a4 e2706e401319fa8ecd2412342e9154 (spike, 1 USDC) and 9f6b8ae96a2d17cf3b a2fd585f1e434287bc7113d33f5ecec75c bbe53470996c (service proof, 0.1 USDC, local co-signer)	B2 and B5
Co-signer topology	Separate cosigner service, same project, private networking, no public domain	cosigner-deployment.md still describes a separate project with a public domain in its diagram and step 3; the amendment note at the top and ADR-0001 describe the deployed topology	Cite the ADR amendment; the diagram is historical
Daily caps	200 USDC on both services (2000000000 units)	Examples use 100 USDC (1000000000 units)	Examples only; live value recorded here
Cold-reserve policy	Not configured on web	10 / 12 / 5 USDC used in the 2026-09-08 proof; 25 / 100 / 50 in the runbook example	B3: set and record actual values
Scheduler and alerts	CRON_SECRET and DISCORD_WEBHOOK_URL absent	Runbooks assume an external scheduler and a Discord webhook exist	B1
Cosigner database variables	COSIGNER_DATABASE_URL and DATABASE_URL both present	Runbook requires COSIGNER_DATABASE_URL only and warns against DATABASE_URL on the cosigner	B4
USDC issuer	GBBD47IF6LWK7P7MDEVSCWR7DPUWV3 NY3DTQEVFL4NAT4AQH3ZLLFLA5	Same	Consistent
Cold reserve account and signers	GDPGRS4P6UZZK23CKKELGLJAYTCAWP V4C7TH6Q322SF735A5H6U5XK5G	Same, in stellar-cold-reserve-runbook.md	Consistent

5. Assessment of issues #5-#13 and #80

Issue state is from GitHub on 2026-09-08. "Readiness for QA" is the developer's view of whether QA can execute the related cases today without a code change or a secret.

Issue	State	Scope	Readiness for QA	Developer work still required
#5	Closed	Multisig payout account (2-of-3, thresholds 2/2/2)	Caveat	B2: point runbooks at the live account; publish current key-custody statement and a redacted secret-scan result.
#6	Closed	Testnet payout spike: friendbot, sponsored trustline, two-signature USDC payment, fee in XLM	Caveat	B6: fresh recipient flow against the live account with expected balance deltas and an evidence template.
#7	Closed	Payout service: exact units, sequence-safe submit, ambiguity reconciliation	Blocked	B6: payout trigger commands, reset steps, concurrency count, ambiguity and persistence-failure controls.
#8	Closed	Independent policy co-signer (same-project service, HMAC transport, read-only ledger role)	Blocked	B4 read-only proof; B5 deployed-path payout; B6 QA-safe tamper/replay harness and access path to the private service.
#9	Closed	Daily payout cap at both signers (200 USDC each, different windows)	Caveat	B6: usage seeding and reset, both bypass directions, breach alert observation (needs B1).
#10	Closed	Hot/cold reserve: 2-of-3 cold account, exact refill ceremony	Blocked	B3 policy on web; B1 scheduler; custodian-run single-signer rejection and two-signer refill with QA observing.
#11	Closed	Dual-asset wallet health, anomaly alerts, cooldown	Blocked	B1 scheduler and webhook; B6 deterministic low-balance, outage, and cooldown controls; admin status view access.
#12	Closed	Payments-lane CI job, no-single-key guard, cap-alert ordering, concurrency	Ready	Attach the payments-lane CI run link for the base SHA to #80. Residual risks are in section 10.
#13	Open	Developer readiness gate (this document)	Blocked	Close B1 to B7, then confirm on #80 that the environment and mechanisms are ready.
#73	Closed	Follow-up fixes and evidence for #5, #7, #10, #11; retry-lease hardening	Ready	None. Its residual risks are carried into section 10.
#80	Open	QA execution and DOCX revision	QA-owned	Blocked by #13. QA revises H1 and H2 wording, then executes once #13 confirms readiness.

6. Test-case mapping D1-TC-001 to D1-TC-028

IDs, titles, and severities match the DOCX exactly. "Readiness" is the developer's current status for that case:

Ready means QA can execute it today from public data or the live site; **Caveat** means it is executable once a specific documented input or control is handed over; **Blocked** means a listed blocker prevents execution.

"Developer provides" names the inputs QA will receive; "Evidence" says what QA records.

ID / Severity	Case	Readiness	Developer provides / QA uses	Evidence QA records	Ref
TC-001 P0	On-chain signer weights and thresholds	Ready	Live account GCP34RITQI . . . ; expected policy 2/2/2 with three weight-1 signers (section 3.1); stellar.expert and Horizon links.	Horizon thresholds and signers JSON; stellar.expert screenshot.	B2
TC-002 P0	Single/master signer cannot authorize payout	Caveat	A developer-run, testnet-only script that builds a valid payment, signs with exactly one key, and submits; expected Horizon result tx_bad_auth. QA never holds a seed; QA observes and records.	Failed submission result code; before/after recipient balance; no new payout tx.	B6, A1
TC-003 P1	Set-options configuration evidence	Ready	Setup tx e966c0a5c27c . . . (3 operations) and the resulting account state.	Transaction link plus current account state.	B2
TC-004 P0	Execute a real two-signature USDC payout	Caveat	Exact payout trigger (withdrawal flow on the site with a labeler account whose balance meets MIN_WITHDRAWAL_UNITS, or the payout-job path), a funded recipient with a USDC trustline, and the expected exact delta. Existing proof: 5083dd72a16b . . .	Public tx hash; signature count on inner and outer envelopes; before/after balances.	B5, B6
TC-005 P1	Centient pays the XLM network fee	Ready	Fee-bump envelope evidence: fee account = payout account, fee charged 200 stroops on the live payout.	Horizon transaction fee_account and fee_charged; recipient XLM unchanged.	-
TC-006 P1	Sponsored or zero-XLM recipient receives USDC	Caveat	Sponsored-recipient flow (/api/me/wallet/sponsor in-app "Set up USDC payouts" step) run against the live account, plus the historical proof in section 3.4.	Before/after XLM and USDC; B6 sponsorship tx; payout tx.	
TC-007 P0	Exact 7-decimal USDC amount construction	Caveat	Payout amounts including a 7-decimal boundary value, the integer unit expected on-chain, and the unit-test reference (payout-amount tests in the payments lane).	Requested units versus Horizon amount for each payout.	B6
TC-008 P0	Idempotent payout request does not double-pay	Caveat	Deterministic replay instructions for one payout reference before and after confirmation, including the admin retry route behaviour (409 while a retry claim is live).	Payout records, tx hashes, recipient balance.	B6
TC-009 P0	Concurrent payouts do not double-settle	Caveat	A concurrency runner with a fixed request count (the automated regression uses 12 simultaneous payouts and requires 12 distinct sequence numbers) and the pass calculation: zero duplicate ledger rows, zero duplicate on-chain transfers.	Runner output, tx hashes, payout ledger rows.	B6
TC-010 P0	Ambiguous Horizon outcome reconciled before retry	Blocked	A testnet-only control that makes the submit response ambiguous while retaining the envelope hash, and the documented reconciliation steps (failure runbook).	Redacted logs showing the hash-based resolution; Horizon lookup of the same hash.	B6

ID / Severity	Case	Readiness	Developer provides / QA uses	Evidence QA records	Ref
TC-011 P0	Post-broadcast persistence failure cannot cause second broadcast	Blocked	A testnet-only control that fails persistence after Stellar acceptance, then a restart/retry procedure that proves no second broadcast.	Original tx proof; recovery logs; recipient balance; single payout row.	B6
TC-012 P1	Permanent rail errors do not enter unsafe retry loop	Caveat	Fixtures: a funded recipient without a USDC trustline (expect op_no_trust) and an unfunded G address (expect op_no_destination); expected behaviour: failed, refunded, retry budget consumed.	Error classification in records/logs; absence of a successful tx.	B6
TC-013 P0	Co-signer independently validates amount and destination	Caveat	A QA-safe harness that sends HMAC-signed requests to the private cosigner service with a tampered destination, amount, or asset. QA never sees the shared secret; the harness reads it from the environment. Expected: 409 with no signature.	Refusal responses; no new payout tx.	B4, B5, B6
TC-014 P0	Non-validated and already-paid payouts are refused	Caveat	Fixtures for a non-validated submission and an already-paid payout job; harness from TC-013. Expected 409 ("no ledger row" / "already carries broadcast hash").	Refusal responses plus chain state.	B6
TC-015 P1	Unauthenticated, stale, replayed, malformed requests fail closed	Caveat	Harness modes: missing signature (401), stale timestamp (401), replayed nonce (401), malformed body (400). The service also refuses with 503 if the isolation level is impermissible.	HTTP status codes and redacted logs.	B6
TC-016 P1	Co-signer database role is read-only	Blocked	Redacted transcript of the runbook permission probes run as the co-signer role: SELECT on submissions and payout_jobs succeeds; UPDATE, DELETE, INSERT, and SELECT on users fail.	The transcript (no connection string).	B4
TC-017 P1	Payout below, at, and above the daily cap	Caveat	Both caps are 200 USDC; usage-seeding and reset steps for the rolling 24 h (app) and UTC-day (cosigner) windows; breach alert observation needs B1.	Cap calculations, both signer decisions, tx presence/absence.	B1, B6
TC-018 P1	Bypassing one cap gate still cannot settle over-cap	Caveat	Direction 1: DAILY_PAYOUT_CAP_UNITS=0 on web, cosigner refuses with 409. Direction 2: generous COSIGNER_DAILY_CAP_UNITS on cosigner, web refuses (0 is invalid there). Both require a redeploy of the changed service.	Refusal at the remaining gate; no on-chain payment.	B6
TC-019 P1	Concurrent cap decisions and residual race	Caveat	Near-cap allowance seeding plus the TC-009 runner. Documented expectation: the cap is a check, not a reservation (section 10, R2), so a bounded overshoot is possible and must be recorded, not hidden.	Concurrent results, aggregate spend, residual-risk note.	B6
TC-020 P1	Cold reserve 2-of-3 and single-signer refill rejection	Caveat	On-chain part is ready (section 3.3). Single-signer attempt is run by one custodian with the refill sign/submit commands; submit refuses fewer than two valid signatures before Horizon is contacted.	Horizon thresholds; failed single-signer result; unchanged balances.	B3, B6
TC-021 P1	Exact refill amount, trigger/target, retained minimum	Blocked	Policy values set on web (B3); POST /api/cron/reserve-refill plan (needs B1); a custodian ceremony QA observes; a violating refill attempt refused at submit.	Refill tx hash; before/after hot and cold balances; runbook reference.	B1, B3

ID / Severity	Case	Readiness	Developer provides / QA uses	Evidence QA records	Ref
TC-022 P1	Low-USDC and low-XLM health states and alerts	Blocked	Public /api/health/wallet already shows the USDC warn state. Low-XLM needs a balance control; alerts need the webhook and cron (B1).	Health responses; Discord alert identity and timestamp.	B1, B6
TC-023 P1	Unavailable health source does not show false healthy	Blocked	A controlled Horizon/asset-source outage (for example an invalid STELLAR_HORIZON_URL on a QA redeploy) and restore procedure. Expected: monitoring status error/unavailable, never healthy.	Health response; alert identity wallet-monitoring-unavailable .	B1, B6
TC-024 P1	Alert cooldown/deduplication and scheduler invocation	Blocked	External scheduler calling POST /api/cron/wallet-health; a short QA-only HEALTH_ALERT_COOLDOWN_MS; delivery results sent then suppressed.	Alert timestamps; scheduler invocation log; two consecutive responses.	B1
TC-025 P0	Repository/CI no-single-key invariant	Ready	The payments-lane CI job for the base SHA, the guard test file, and the local command <code>npm run test:payments</code> .	CI run link; guard test output.	-
TC-026 P1	Cap alert only after required persistence order	Ready	The cap-alert-ordering regression test in the payments lane; optional manual observation needs B1.	Test output for the SHA.	B1 (manual half)
TC-027 P1	Evidence package complete, redacted, SHA-bound	Blocked	This guide, the updated runbooks (B2), the CI link, and the redacted index in section 13.	Completed checklist in section 12.	B2, B7
TC-028 P0	Record final QA verdict	QA-owned	Nothing beyond the workflow in section 12. The verdict is the QA: PASSED label on #80.	Label applied, or failed/blocked IDs recorded on #80.	H1

7. Developer-owned fixtures, reset, access, and fault-injection requirements

This section is the contract behind blocker B6. Each control must be testnet-only, access-controlled, unavailable in the public-network configuration, and described with an exact command or endpoint before #80 begins. QA must never need a seed, a superuser role, a code change, or mainnet access.

7.1 Access QA receives

- Site login: a QA labeler account and a QA customer account on <https://centient.work>, credentials delivered out of band, never in #80 or in evidence.
- Admin status view: an admin login for /admin (the protected area and GET /api/admin/health) showing both balances, reserve state, and degraded sources.
- Public health: GET /api/health/wallet (no auth).
- Cron routes: POST /api/cron/wallet-health, POST /api/cron/payout-retry, POST /api/cron/payout-reconcile, POST /api/cron/reserve-refill - all require Authorization: Bearer <CRON_SECRET>. The developer either runs them for QA on request or provides a wrapper that injects the secret without displaying it.
- Co-signer: it has no public domain. The developer provides a harness (script or internal route) that runs inside the project network and sends HMAC-signed requests on QA's behalf, with switchable negative modes (see TC-013 to TC-015).
- Logs: redacted log excerpts on request, or read access to the Railway log view for the web and cosigner services if the workspace owner grants it.

7.2 Deterministic fixtures

Fixture	Purpose	Cases
Validated payout reference (submission and payout job kinds)	Positive co-sign and payout	TC-004, TC-007, TC-008, TC-013
Non-validated submission	Co-signer refusal	TC-014
Already-paid payout job (has a broadcast hash)	Co-signer refusal and replay	TC-008, TC-014
Pending, failed, and reconciliation-required payouts	Retry and reconciliation paths	TC-010, TC-011, TC-012
Recipient with a USDC trustline	Happy path	TC-004, TC-009
Recipient without a trustline (funded)	op_no_trust	TC-012
Unfunded G address	op_no_destination	TC-012
Zero-XLM recipient needing sponsorship	Sponsored trustline	TC-006
Labeler account with balance above the minimum withdrawal	Withdrawal trigger	TC-004, TC-017

7.3 Reset and cleanup rules

- Reset only application rows that never reached Horizon. A payout row that carries a transaction hash is never deleted or re-used; it stays visible so the run cannot conceal a broadcast.
- Cap usage is reset by waiting out the window or by seeding a fresh QA window, never by deleting broadcast rows.
- Every reset step is logged in #80 with the time and the rows touched (by id, not by content).
- Balances are restored to a documented baseline after balance-manipulation cases (TC-022, TC-021).

7.4 Fault-injection and control catalogue

Control	Required behaviour	Cases
Single-signer submit	Sign with exactly one payout-account key and submit; Horizon must answer tx_bad_auth. Developer-run; QA observes.	TC-002
Concurrency runner	Fixed N simultaneous payouts (N = 12 to match the regression). Reports sequence numbers, hashes, and ledger rows.	TC-009, TC-019
Ambiguous Horizon response	Drop or delay the submit response after acceptance while the envelope hash is retained; the service must resolve by hash.	TC-010
Persistence failure after acceptance	Fail the broadcast-tuple write after Stellar acceptance; restart; prove one broadcast. Expect alert identity payout-persistence-unavailable.	TC-011
Replay of a payout reference	Repeat the same reference before and after confirmation; admin retry answers 409 while a claim is live.	TC-008
Cap usage seeding	Place rolling and UTC-day usage just below, at, and above 200 USDC without new on-chain spend where possible.	TC-017, TC-019
One-gate bypass	Set DAILY_PAYOUT_CAP_UNITS=0 on web, or raise COSIGNER_DAILY_CAP_UNITS on cosigner; redeploy the changed service only.	TC-018
Balance controls	Move USDC or XLM to cross warn/page thresholds and restore afterwards.	TC-022
Health-source outage	Make Horizon or the asset lookup unavailable for the health check and restore.	TC-023
Cooldown control	Short QA-only HEALTH_ALERT_COOLDOWN_MS, then restore the default.	TC-024
Boot-conflict test	Show the web service refusing to start when COSIGNER_URL and STELLAR_POLICY_SIGNER_SECRET are both set (redacted log line only).	#8 checks in #80
Public-network refusal	Show same-workspace refused when STELLAR_NETWORK=public (unit evidence or a non-deployed dry run; never a live mainnet config).	#8 checks in #80
Insufficient-XLM payout	Drive spendable XLM below the fee floor, attempt a payout, expect a clean retryable failure and an XLM page.	#11 negative scenario in #80

8. Co-signer topology and credential separation

8.1 Deployed topology (ADR-0001 as amended 2026-09-08)

Placement	cosigner service inside the centient-work Railway project, next to web and Postgres
Transport	HTTPS over same-project private networking; HMAC-SHA256 over the raw body with timestamp and nonce replay window, constant-time comparison
Public exposure	None. There is no public domain for the cosigner.
Isolation level	COSIGNER_ISOLATION_LEVEL=same-workspace - refused when STELLAR_NETWORK=public (checked per request, HTTP 503)
Signing key location	STELLAR_POLICY_SIGNER_SECRET exists only on the cosigner service. The web service holds STELLAR_POLICY_SIGNER_PUBLIC to verify the returned detached signature.
Boot guard	The web service refuses to start if it sees both COSIGNER_URL and STELLAR_POLICY_SIGNER_SECRET.
Ledger re-derivation	The cosigner reads the submission or payout job named by the reference over its own read-only role and refuses if validation state, destination, or exact units disagree.
Replicas	Exactly one. Replay nonces and in-flight cap commitments are process-local.
Health	GET /health returns {"status":"ok", "isolation":"same-workspace"} (internal only).

8.2 Refusal codes QA will see through the harness

HTTP	Cause	Cases
401	Missing or invalid HMAC signature, stale timestamp, replayed nonce	TC-015
503	Isolation level not permitted for the configured network	#8 public-network check
400	Malformed or incomplete request body	TC-015
409	Envelope does not match the request; ledger row missing, not validated, or already broadcast; co-signer daily cap reached	TC-013, TC-014, TC-017, TC-018
200	Detached signature returned with the policy public key	TC-004, TC-013 (control)

8.3 Credential-separation requirements

Requirement	Status on the live deployment	Action
Policy seed absent from web	Expected by design; the boot guard enforces it	Developer confirms with a redacted variable-name listing (names only).
Co-signer database role is SELECT-only on submissions and payout_jobs	Unproven (B4)	Attach the probe transcript.
DATABASE_URL absent from cosigner	Not met: both COSIGNER_DATABASE_URL and DATABASE_URL present	Remove or justify; if it is a Railway shared reference, document why the process ignores it.
Master and ops credentials on web	Both present (accepted decision A1)	Record as residual risk R7; no action for the testnet MVP.
Shared HMAC secret identical on both services	Implied by the successful live payout if B5 confirms the remote path	Confirm under B5.
Signing key not in the repository	Guard test scans lib/, app/, services/, scripts/ for secret readers	Cite the CI run for the SHA.

9. Daily cap, reserve, wallet health, alerts, scheduler, and monitoring

9.1 Daily payout cap

Gate	Variable	Live value	Window	Disabled value	Breach behaviour
Payout service (web)	DAILY_PAYOUT_CAP_UNITS	2000000000 (200 USDC)	Rolling 24 h of broadcasts	0	PayoutCapError before anything is built; cap alert path runs. Withdrawal: terminal, refunded, budget consumed. Legacy per-submission job: deferred, budget untouched.
Policy co-signer	COSIGNER_DAILY_CAP_UNITS	2000000000 (200 USDC)	UTC calendar day plus signed-but-unseen commitments	None (must be positive)	HTTP 409, no signature; the two-signature payout cannot be submitted.

- Units: 1 USDC = 10000000 units. 200 USDC = 2000000000 units. Never write a decimal point or an XLM stroop value.
- Changing either cap is a Railway variable change plus a redeploy of that service; no code change.
- The windows differ on purpose; a payout settles only when both windows have room.
- Deferred legacy payouts recover only through POST /api/cron/payout - retry, which needs the scheduler (B1).

- The cap alert reports the ledger's spend plus a "blocked attempt" line for the refused amount.

9.2 Hot/cold reserve

- The deployed application stores only public cold identities and the three policy unit values; no cold seed exists in any deployed service.
- Detection: `POST /api/cron/reserve-refill` answers 200 healthy, 202 refill_required with the exact amountUnits, 503 insufficient_reserve, or 500.
- Ceremony: `npm run stellar:reserve:refill -- prepare` on an online host, then `sign` once per custodian with exactly one seed on an offline host, then `submit` once with `STELLAR_RESERVE_REFILL_AMOUNT_UNITS` set. Submit rechecks balances, shape, fee (max 10000 stroops), time bounds (max 15 minutes), and requires two valid distinct signatures.
- Worst-case loss from a full hot-wallet compromise equals the hot-float target, because every refill restores exactly the target. The developer records the chosen target and the measured daily budget in #80 once B3 is set.
- Historical proof values: trigger 10 / target 12 / retained 5 USDC; refill 3.1 USDC; hot 8.9 to 12.0 USDC; cold 10.0 to 6.9 USDC.

9.3 Wallet health and alerts

Signal	Variable (default)	Meaning
USDC warn / page	BALANCE_WARN_USDC (50) / BALANCE_PAGE_USDC (10)	Float that funds payouts
XLM warn / page	BALANCE_WARN_XLM (5) / BALANCE_PAGE_XLM (2)	Spendable XLM after reserves and sponsored reserves; funds every fee
Payout rate spike	HEALTH_PAYOUT_COUNT_THRESHOLD (100) in HEALTH_PAYOUT_WINDOW_MINUTES (60)	Anomalous payout count
Payout volume spike	HEALTH_PAYOUT_VOLUME_UNITS_THRESHOLD (1000000000)	Anomalous payout volume
Repeated failures	HEALTH_FAILURE_COUNT_THRESHOLD (3) in HEALTH_FAILURE_WINDOW_MINUTES (15)	Permanent payout failures
Cap threshold	HEALTH_CAP_PERCENT_THRESHOLD (80)	Percent of the rolling cap consumed
Refill overdue	HEALTH_REFILL_OVERDUE_MINUTES (30)	Refill required but not executed
Cooldown	HEALTH_ALERT_COOLDOWN_MS (900000 = 15 min)	Shared by every alert identity via Redis

Alert identities: wallet-usdc-warn, wallet-usdc-page, wallet-xlm-warn, wallet-xlm-page, payout-rate-spike, payout-volume-spike, payout-cap, repeated-payout-failures, reserve-refill-overdue, plus the monitoring-source failures wallet-monitoring-unconfigured, wallet-monitoring-unavailable, payout-monitoring-unavailable, reserve-monitoring-unconfigured, reserve-monitoring-unavailable, refill-timer-unavailable, health-monitor-unavailable, and payout-persistence-unavailable. A missing metric is never read as healthy.

Delivery results: sent, suppressed, sent-degraded, suppressed-degraded, failed, disabled. On the live deployment every result is currently disabled because `DISCORD_WEBHOOK_URL` is unset (B1). PAGE alerts still deliver when Redis is down; WARN alerts fail closed.

9.4 Scheduler

- Nothing in the repository schedules the cron routes; `railway.json` only runs migrations before deploy.
- Required: an external scheduler with `CRON_SECRET` in its own secret store calling `POST /api/cron/wallet-health` at least once per minute, `POST /api/cron/payout-retry` and `POST /api/cron/payout-reconcile` on a short interval, and `POST /api/cron/reserve-refill` hourly.
- QA proves scheduler invocation (TC-024) from the scheduler's own run log plus matching timestamps in the health responses; the developer provides both.

10. Residual risks carried from #12, #73, and ADR-0001

The final QA record on #80 must state these without overstating guarantees. They are not defects to be fixed during QA; any observed behaviour outside the stated bound is a defect.

ID	Risk	Bound and backstop	Where QA meets it
R1	Co-signer isolation is a separate service in the same Railway project, workspace, account, control plane, repository, CI, and database instance. It protects against a compromised application container, not against a compromised Railway account or collaborator.	same-workspace fails closed on the public network; mainnet requires a separate account (ADR-0001 exit criteria).	TC-013 to TC-016; record the isolation level.
R2	The daily cap is a check, not a reservation. Concurrent decisions can each pass and together exceed the cap.	Bounded by the cap alert and health monitoring; no payout settles twice. Asserted as behaviour in the concurrency regression.	TC-019
R3	Cap alerts are fire-and-forget at the three persistence sites; a Redis or Discord failure loses that individual alert.	The health monitor raises the same alert identity from the ledger.	TC-026, TC-024
R4	Process-death window: the envelope hash lives in memory between submit and resolution. A crash there loses the identity and the job requeues without it.	The retry lease is refreshed while the broadcast is in flight; the failure runbook routes the case to manual reconciliation before any re-run.	TC-010, TC-011
R5	The submit mutex is process-local; the deployment must run exactly one payout submitter.	A second submitter costs tx_bad_seq and a rebuild, not a duplicate; horizontal scaling needs a distributed lock.	TC-009
R6	The retry lease is conservative and is not a fencing token. lastRetriedAt cannot distinguish an in-flight retry from a recently completed failed retry, so the admin retry route answers 409 while a claim is live.	Cron stands down on a live lease; a degraded process can still be reclaimed mid-payout.	TC-008
R7	Accepted decision A1: the web service holds both the master and ops credentials for the payout account, so one compromised web process could produce two of three signatures.	Testnet only, zero real value at risk; the on-chain 2-of-3 still refuses any single key; the co-signer path still validates every payout it signs.	TC-002; record with R1.

11. Safe evidence and redaction rules

Never record: seed phrases or S... secrets, private keys of any kind (including the legacy PAYOUT_PRIVATE_KEY), HMAC values or COSIGNER_SHARED_SECRET, CRON_SECRET, database URLs or passwords, Discord webhook URLs, admin or session tokens, signed or unsigned pending refill XDRs, raw environment listings, or full unredacted logs.

Safe to record

- Public G... account identifiers, transaction hashes, ledger numbers, stellar.expert and Horizon links.
- Variable names (never values) and non-secret configuration values such as cap units and thresholds.
- HTTP status codes, alert identities, delivery results, timestamps, and error class names.
- Balances before and after, exact unit amounts, fee charged, signature counts.
- Test IDs, the full commit SHA, the build URL, tester, and date.
- Screenshots of stellar.expert, the public health endpoint, the admin status view, and Discord alert messages with any webhook path cropped out.

How to redact

- Replace any secret with the literal text [REDACTED] and keep the variable name so the context remains verifiable.
- Crop or mask URLs that carry tokens, including Discord webhook paths and database hosts with credentials.
- Before attaching a log excerpt, search it for S (56 characters), 0x followed by 64 hex characters, postgresql://, and discord.com/api/webhooks; remove any match.
- Do not attach terminal recordings or environment listings; attach the specific lines that prove the case.

Evidence locations

- Executed test record and evidence index: attached to or linked from #80.
- Public chain evidence: stellar.expert links in section 3 and 13.
- Automated evidence: the payments-lane CI job for fa36cf4663fe on GitHub Actions.
- Runbooks: docs/ in the repository at the base SHA (section 13), updated under B2.

12. Final handoff checklist and the QA:PASSED workflow

12.1 Developer handoff checklist (close on #13 before #80 begins)

Done	Item	Blocker
[]	Set CRON_SECRET and DISCORD_WEBHOOK_URL on web; provision the external scheduler; give QA the test Discord channel.	B1
[]	Update the multisig, spike, and payout-service runbooks to the live account or label the old sections historical.	B2
[]	Set the cold-reserve public identities and policy unit values on web; confirm the reserve-refill plan responds.	B3
[]	Attach the redacted co-signer role permission-probe transcript; remove or justify DATABASE_URL on cosigner.	B4
[]	Record the provenance of the live payout's second signature or run one payout through the deployed cosigner.	B5
[]	Deliver every control in section 7.4 with exact commands, plus the fixtures in 7.2 and access in 7.1.	B6
[]	Rotate or remove the legacy PAYOUT_PRIVATE_KEY with owner authorization.	B7
[]	Top up the hot float from the faucet or reserve and record the balances.	-
[]	Attach the payments-lane CI run link for the base SHA to #80.	TC-025
[]	Confirm on #80: build URL, full SHA, deployment timestamp, and that all mechanisms are ready. Hold environment-affecting changes during the run.	#13 acceptance

12.2 QA workflow on #80 (simplified)

- 1 Revise the DOCX wording (H1, H2) so IDs stay identical and every case names its developer-provided control. Attach or link the revised document on #80.
- 2 Wait for #13 to confirm readiness on #80.
- 3 Execute every applicable case against the single environment and the full SHA. Record Actual Result, Status (Pass, Fail, Blocked, Not Tested), evidence or bug ID, tester, date, build URL, and SHA. A blocked case is not a pass.
- 4 Record the residual risks R1 to R7 in the execution summary without overstating guarantees.
- 5 Confirm the pass conditions: no case remains Not Tested or Blocked; every P0/P1 case passes; no unresolved P0/P1 Deliverable 1 defect; evidence is redacted and consistent; no duplicate or unreconciled payout remains.
- 6 When all conditions pass, apply the QA: PASSED label to #80. That label is the sole approval signal.
- 7 If any condition fails, do not apply the label; record failed or blocked IDs and linked defect issues on #80.

Not required from QA: a QA PASSED <full-sha> comment, a Claude review link, closing #13 or any implementation issue, or updating Epic #4. Those belong to the developer and maintainer. #13 is marked complete by the developer only after blockers B1 to B7 are actually resolved.

13. Public and reference links

13.1 GitHub

Reference	Link
Issue #13 (developer readiness)	https://github.com/webnxt-2030/Centient/issues/13
Issue #80 (QA execution)	https://github.com/webnxt-2030/Centient/issues/80
Epic #4	https://github.com/webnxt-2030/Centient/issues/4
Issues #5 to #12	#5 , #6 , #7 , #8 , #9 , #10 , #11 , #12
Issue #73 (follow-up fixes)	https://github.com/webnxt-2030/Centient/issues/73
Base commit	fa36cf4663fecea6137dcd4e73b136baa49d34f0
CI workflow	https://github.com/webnxt-2030/Centient/actions/workflows/ci.yml

13.2 Repository documents (develop branch)

Document	Path
Multisig runbook (#5)	docs/stellar-multisig-runbook.md
Payout spike (#6)	docs/stellar-multisig-payout-spike.md
Payout service (#7)	docs/stellar-multisig-payout-service.md
Cold-reserve runbook (#10, #11)	docs/stellar-cold-reserve-runbook.md
Daily payout cap runbook (#9)	docs/stellar-daily-payout-cap-runbook.md
Payout failure runbook	docs/stellar-payout-failure-runbook.md
Co-signer deployment (#8)	docs/cosigner-deployment.md
Payments-lane evidence (#12)	docs/payments-lane-evidence.md
ADR-0001 simulated co-signer isolation	docs/adr/0001-simulated-cosigner-isolation.md
Environment variable names	.env.local.example
No-single-key guard test	lib/stellar/_tests_/no-single-key-payout.test.ts
Concurrency regression	lib/_tests_/payout-concurrency-db.test.ts
Cap-alert ordering regression	lib/_tests_/payout-cap-alert-ordering-db.test.ts
Cap independence tests	lib/_tests_/payout-cap.test.ts and lib/stellar/_tests_/cosigner-service.test.ts
Payments-lane definition	tests/payments-lane.ts

13.3 Stellar Testnet

Item	Link
Live payout account	https://stellar.expert/explorer/testnet/account/GCP34RITQIVSLHS5T4XZRENIBUS3T7FHL3VSR24GK7HPMHGAACKWK4BUO
Live set-options tx	https://stellar.expert/explorer/testnet/tx/e966c0a5c27cbe0253f2812d158b38f8e91513de48254923f9ecb6f4c19630fd
Live 1 USDC payout tx	https://stellar.expert/explorer/testnet/tx/5083dd72a16bfa749c6b302c293e931939acd70cc52f63586443921d60698206
USDC issuer	https://stellar.expert/explorer/testnet/account/GBBD47IF6LWK7P7MDEVSCWR7DPUWV3NY3DTQEVFL4NAT4AQH3ZLLFLA5
Cold reserve account	https://stellar.expert/explorer/testnet/account/GDPGRS4P6UZZK23CKKELGLJAYTCAWPV4C7TH6Q322SF735A5H6U5XK5G
Cold trustline / setup / funding / refill	https://stellar.expert/explorer/testnet/tx/dddea73e981194f020bcf531f96140c3117f8c400915228e6901c2312b4a8c6a https://stellar.expert/explorer/testnet/tx/05bbe397033a7984700b8d844cf1d247e04db7537fd9a0293a5bda6c1506d14a https://stellar.expert/explorer/testnet/tx/452fd68061ecae052ebd681ee47010adbc2e05c01c86afb9bba685a77fb1d836 https://stellar.expert/explorer/testnet/tx/3a5969cdac22dad6646630c90cdb3ae3919a727f2abd8f663863b7e9e6b9ef3e
Historical payout account	https://stellar.expert/explorer/testnet/account/GC5UOKLU6J2EROZYYP2I23ZEF4YF42TGGRNQMGTGOJGJ7NOCH3TTR4A6
Historical set-options / sponsor / spike / service proof	https://stellar.expert/explorer/testnet/tx/8a16d6236cbb0aa1887eba35f6b4b04e41e289532df06b1d13413551e8fc1ad5 https://stellar.expert/explorer/testnet/tx/63dba35261787b2fa05c0c1c833a02f5983c3a4e229d984a7ed1ecf3c881a71b https://stellar.expert/explorer/testnet/tx/be8a6a2363401ff7f3634b8af47db88a4e2706e401319fa8ecd2412342e9154 https://stellar.expert/explorer/testnet/tx/9f6b8ae96a2d17cf3ba2fd585f1e434287bc7113d33f5ecec75cbbe53470996c
Sponsored recipient	https://stellar.expert/explorer/testnet/account/GA4XSHGIOVGWI6ZGDIRFG3ROFOHXRDVHH5YKGFKC5C2LHOT6EB5WCQXI
Horizon testnet	https://horizon-testnet.stellar.org
Public wallet health	https://centient.work/api/health/wallet
Circle testnet faucet	https://faucet.circle.com/?allow=true
Stellar fee-bump transactions	https://developers.stellar.org/docs/learn/fundamentals/transactions/fee-bump-transactions
Stellar sponsored reserves	https://developers.stellar.org/docs/learn/encyclopedia/sponsored-reserves

End of document. Version 1.0, 2026-09-08. Prepared by the Centient developer team for issue #13. Stellar Testnet only.